HIREME — ELITE INTERVIEW COACHING DOSSIER

Target Role: Quantitative Analyst (Fund Discovery & Factor Intelligence)
Employer: PortfolioFuture (New York, NY · Remote) | **Candidate:** Bhupesh Chandra Dimri

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STEP 1: SKEPTICAL SENIOR RECRUITER AUDIT

Category	Recruiter Evaluation & Strategic Counter-Positioning
Top 3 Selling Points	1. Direct mathematical alignment: built complete return decomposition & factor exposure engines. 2. Institutional data fluency: hands-on extraction & synthesis of SEC Form 13F & Form ADV filings. 3. Proven commercial orientation: framed algorithms around saving RIAs $140.8M in fee drag.
Top 3 Weak Spots / Flags	1. Traditional finance credential vs pure Math/Physics PhD perception. 2. Internship title vs senior-level modeling depth. 3. Remote accountability and autonomous delivery pace.
Hidden Advantages	You built an actual live repository (PortfolioFuture-SubstAI) solving their exact business problem (10,000+ funds, 130M+ relationships, pairwise substitution) before even stepping into the first round.
Likely Objections	"Can this candidate write production Python that scales across 10,000 tickers daily, or do they only run static notebooks?" → Proof: Modular src/ pipeline with vectorized Polars/DuckDB execution.

STEP 2: POSITIONING STRATEGY & NARRATIVE SCRIPTS

The 60-Second Hook: 'Tell Me About Yourself'

"I'm a quantitative finance practitioner specializing in empirical asset discovery, factor return decomposition, and institutional portfolio intelligence. Over the past several years, I've focused on bridge-building between complex financial econometrics—like Fama-French multi-factor modeling and active share analysis—and scalable Python architectures that parse regulatory datasets like SEC Form 13F and Form ADV. Most recently, seeing PortfolioFuture's mission to evaluate 10,000+ funds across 130 million relationships, I engineered 'PortfolioFuture-SubstAI'—a modular Machine Learning framework that decomposes fund returns, measures pairwise cosine similarity, and identifies fee-advantaged ETF substitutes, mapping over $140M in potential RIA fee savings across $26B in analyzed capital. I'm here today because PortfolioFuture sits at the exact frontier of quantitative transparency and commercial advisor discovery, and I'm eager to bring that same engineering rigor and domain ownership to your research team from day one."

Three Tailored STAR Narratives

Theme	STAR Story Arc (Quant Analyst Calibration)
1. Technical Rigor (Factor Modeling)	**S:** Legacy fund evaluations relied on nominal 3-year returns, masking high-beta market risk as managerial skill. **T:** Architect a multi-factor econometric engine isolating true alpha from Fama-French risk factors. **A:** Built a rolling 36-month OLS & Ridge regression pipeline with VIF collinearity pruning. **R:** Revealed that 64% of active equity funds charged active fees for synthetic market beta.

2. Scale & Data Engineering (SEC 13F)	**S:** Institutional RIA ownership data was fragmented across thousands of quarterly XML/EDGAR filings. **T:** Create an automated ingestion pipeline mapping $26.8B in position capital across 500+ wealth managers. **A:** Leveraged DuckDB and vectorized Polars to parse, normalize, and match CUSIPs with fund share classes. **R:** Compressed monthly ETL run-times from 6 hours to under 4 minutes with 100% relational integrity.
3. Commercial Impact (Fee Drag)	**S:** Financial advisors lacked objective mathematical evidence to justify mutual fund-to-ETF transitions. **T:** Build a pairwise substitution algorithm satisfying strict tracking error and factor alignment thresholds. **A:** Trained a Random Forest classifier achieving 100% precision with tracking error $\leq$ 2.50% and fee savings $\geq$ 15 bps. **R:** Unlocked $140.8M in annualized addressable client savings, creating an undeniable sales catalyst for advisors.

STEP 3: HIGH-STAKES INTERVIEW QUESTIONS & WINNING RESPONSES

Q1: 'How do you handle multi-collinearity when regressing fund returns against Fama-French 5-factor benchmarks?'

Winning Response: "When regressing fund excess returns against Mkt-RF, SMB, HML, RMW, and CMA, correlation between size and investment factors can inflate variance inflation factors (VIF). In my workflow, I first compute condition indices and inspect VIF across the factor correlation matrix. If collinearity exceeds threshold (VIF > 5), I utilize an ElasticNet regularization pipeline with cross-validated L1/L2 penalties, which stabilizes coefficient variance while preserving economic interpretability. Furthermore, for rolling factor betas, I apply exponentially weighted moving windows to prevent sudden step-changes from regime shifts."

Q2: 'What is your methodology for defining a viable ETF substitute for an established mutual fund?'

Winning Response: "A credible substitution must satisfy three simultaneous hurdles: First, mathematical similarity—a rolling 36-month return cosine similarity ≥ 0.95 and an annualized tracking error $\leq 2.50\%$. Second, structural factor parity—ensuring the substitute does not inadvertently alter the client's factor exposure profile (e.g. drifting from Value to Growth). Third, economic hurdle—an expense ratio compression of at least 15 to 25 basis points with sufficient secondary market liquidity (AUM > $100M, median bid-ask spread < 0.08%). In my SubstAI model, this is framed as a supervised classification problem where precision is prioritized at 100% to ensure zero false-positive recommendations reach advisors."

Q3: 'How do you parse and reconcile messy SEC Form 13F filings across varying share classes and CUSIP renamings?'

Winning Response: "SEC 13F XML files only report CUSIPs, issuer names, and aggregate values—not fund tickers or share class distinctions. I construct a dual-layer reconciliation pipeline: first, a master entity cross-reference mapping CUSIPs against SEC Form N-PORT and CRSP fund identifiers; second, a fuzzy matching and corporate action resolver that handles CUSIP changes, fund reorganizations, and liquidations. We store these mappings in an immutable DuckDB ledger with automated daily sanity checks against total institutional float."

STEP 4: REVERSE-INTERVIEW QUESTIONS (TOP 1% CANDIDATE)

1. *"When PortfolioFuture evaluates fund substitution across 130M+ relationships, how do you dynamically balance tracking error tolerance against tax-drag implications for taxable advisory accounts versus retirement accounts?"*
2. *"How is your quantitative research team currently isolating active manager skill during macro regime transitions where traditional Fama-French factor correlations break down?"*
3. *"What is the internal roadmap for expanding your discovery engine into private wealth alternatives or custom direct indexing models?"*

STEP 5: POST-INTERVIEW HIGH-CONVERSION FOLLOW-UP

Subject: Thank you & follow-up: Empirical fund substitution & SEC 13F analytics

"Hi [Interviewer Name],

Thank you for the inspiring conversation today regarding PortfolioFuture's mission to bring empirical rigor to fund discovery across 20,000+ advisory firms.

Our discussion regarding the balance between factor beta decomposition and tracking error in fund substitution resonated strongly with me. As mentioned, I've published a working implementation of this workflow—decomposing factor exposures and analyzing $26.8B in SEC 13F RIA capital—in my open-source repository: github.com/bhupesh-debug/ml-project-portfoliofuture-substai.

I'm even more enthusiastic about the opportunity to contribute to PortfolioFuture's research velocity. Looking forward to the next steps!

Best regards,
Bhupesh Chandra Dimri"